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Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation (FPDAF)

23CSE399 – Project Phase 1 (Technical Progress Report - Review 2)
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1 Project Proposal and objectives

Overall Project Idea: Instead of merely implementing standard off-the-shelf algorithms, this project proposes a novel mathematical extension to existing FedAvg and FedPer models. We introduce the fully decentralized Federated Personalized Drift-Aware Attention Framework (FPDAF) specifically designed for multivariate healthcare time-series analysis (e.g., ICU risk forecasting).

Primary Novelty & Algorithmic Innovation: The definitive novelty of FPDAF lies in its dynamic hybrid integration. Unlike conventional FL architectures that maintain blind static global averaging, our extension mathematically intercepts the FedAvg loop. We inject a CUSUM Statistical Trigger directly into local active hospital nodes, enabling isolated Neural Network layers (Personalization Heads) to mathematically detach and adapt in real-time when patient physiology rapidly deteriorates, utilizing Temporal Attention to explain exactly which vitals caused the deviation.

Objectives to Achieve:

1. Enable cross-hospital predictive modeling on patient data without violating HIPAA/GDPR privacy constraints.
2. Manage high patient heterogeneity across hospitals using explicit Local Personalization Heads.
3. Detect when a patient's core physiological state changes using CUSUM Drift Detection algorithms to automatically retrain and dynamically adapt models locally.
4. Integrate Temporal Attention weights to provide clinicians with transparent insights into exactly what historical vital signs caused a risk prediction.

2 Introduction and Background

Continuous monitoring systems in Intensive Care Units (ICUs) collect immense streams of physiological data (HR, BP, SpO2). Early detection of patient deterioration using Deep Learning models can prevent fatalities like Sepsis or Cardiac Arrest. However, massive clinical models require centralized data, violating privacy laws. While Federated Learning (e.g., FedAvg) resolves the data-sharing issue, it fails to account for non-IID data distributions (how demographics and hospital protocols vary significantly between facilities).

Our literature review explored standard Federated methods. FedAvg (McMahan 2017) trains efficiently but lacks personalization. FedPer (Arivazhagan 2020) personalizes models to local distributions but lacks any mathematical mechanism for tracking concept drift over time. FedXAI works

(Kalakoti 2025) focus mostly on classification datasets, not complex high-frequency critical care time-series. Thus, a significant gap exists in achieving privacy, personalization, drift monitoring, and clinical explainability simultaneously.

3 Execution plan

The subdomains required to solve this problem involve heavily modifying base components: Data Engineering (custom structural sequence profiling), Deep Learning Architecture (extending LSTMs with novel Temporal Attention constraints), Decentralized Systems (modifying generic FedAvg aggregation to include attention weights), Model Customization (dynamically manipulating FedPer baseline layers), and Statistical Math (altering standard CUSUM tracking into an active Neural Network layer-freezing trigger).

Our plan mapped these subdomains to execution steps: defining the novel mathematical intersection of these bases in Phase 1, followed by developing the PyTorch dataset pipeline, establishing the generic baseline variants locally, and finally attaching our novel drift-detector and attention overlays structurally in Python to prove the FPDAF extension in Phase 2.

4 Fundamentals which you have learned in the semester

To prepare for implementation, the team completed an extensive curriculum covering the fundamental requirements:

- Textbooks & Papers: Thoroughly studied the foundational architectures of Federated Learning (FedAvg, pFL, and FedPer) to mathematically understand secure aggregation.
- PyTorch Engineering: Used MOOCs and official documentation to learn how to explicitly sever Neural Network graphs in PyTorch (allowing us to separate "Global Backbones" from isolated "Local Heads") and simulate custom training loops iteratively without data centralization.
- Statistical Mechanics: Learned the equations and thresholds defining the Cumulative Sum (CUSUM) drift detection standard to track temporal shifts in residual streams.

5 Architecture diagram and explanation of the design

The FPDAF architecture is a split-layer design. The Global Temporal Backbone consists of the LSTM layers responsible for learning generalized physiological laws across all participating hospitals.

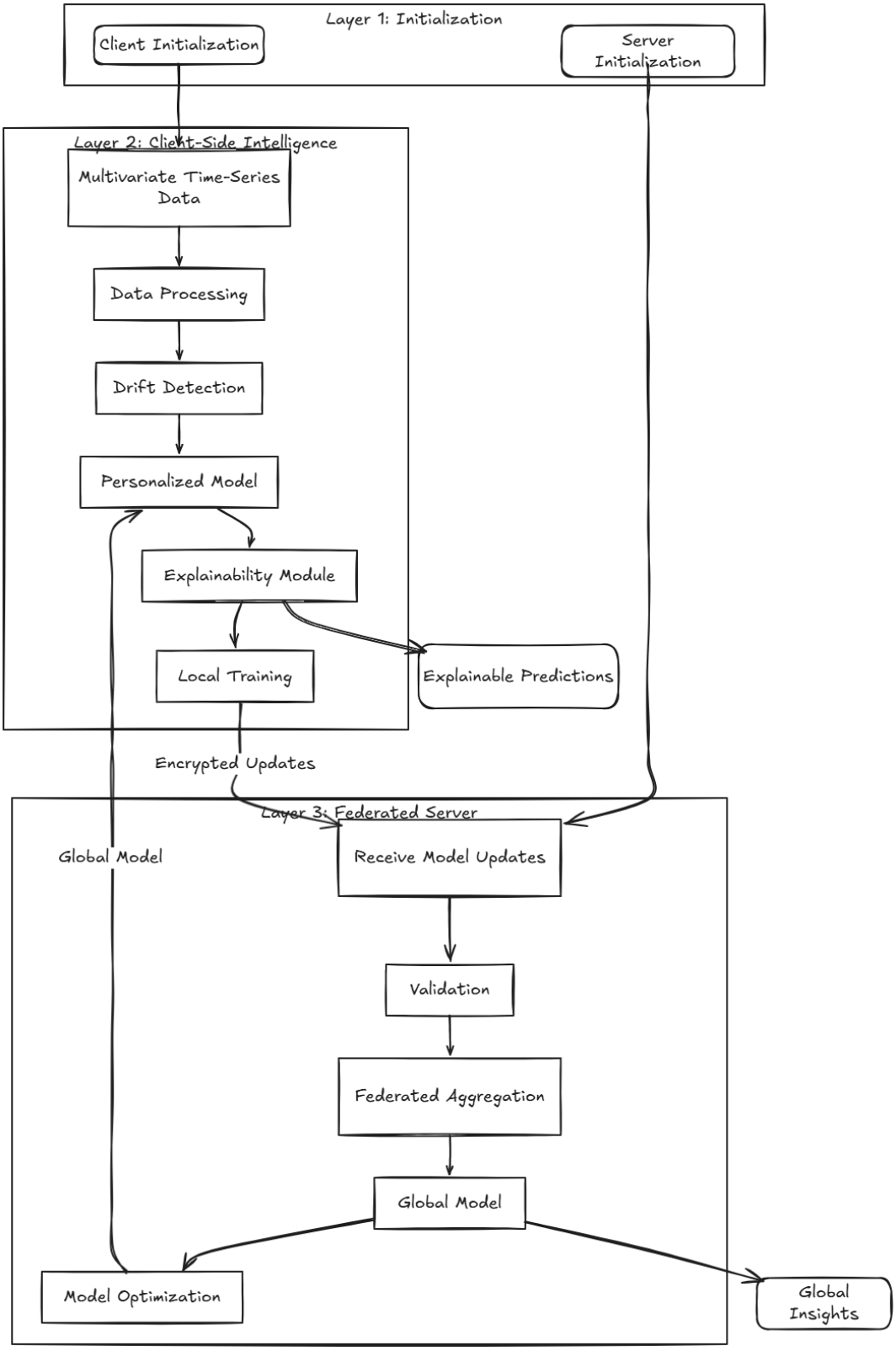

Figure 1: System Architecture highlighting separation of Global Backbone (server-synced) and Local Heads (hospital-isolated).

The Attention Layer sits atop the backbone to compute temporal context vectors based on recent physiological spikes, giving transparent x_i timestamps feature importance. Finally, the Personalization Head is isolated completely to the hospital node; it never transfers its gradients to the server, ensuring the final clinical decision is strictly correlated to the local hospital's demographic

distribution.

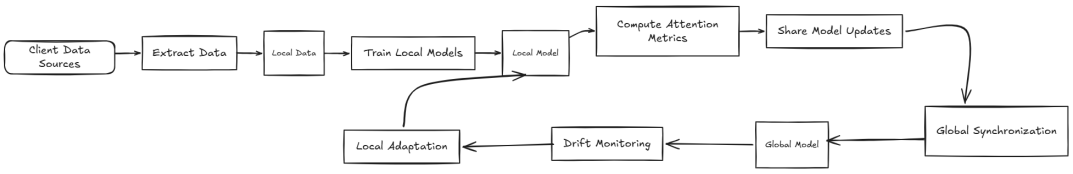

Figure 2: FPDAF System Dataflow Mapping - Illustrating end-to-end data processing.

6 Summary of works completed / research findings (Technical Evidence)

During this semester (Phase 1), the following milestones were completed mapping to our Technical Validation & Evidence requirements:

1. Domain Consolidation: Successfully shifted use-case entirely into ICU Healthcare datasets.
2. Mathematical Definition: Derived the CUSUM threshold trigger mechanism $S_t = \max(0, S_{t-1} + (e_t - \mu - k))$ specifically for linking to local model resets.
3. Architecture Creation: Finalized the fully decentralized, visually mapped FPDAF blueprint.
4. PyTorch Baseline Simulation Setup: Created the GitHub repository (https://github.com/aksharsakhi/23CSE399-Project-Phase-1), synced a foundational FedAvg-PyTorch implementation baseline, mocked a SimulatedICUDataset class, and began structurally coding the decoupled model heads in Python.

7 Plans for the oncoming semesters

Having successfully acquired the PhysioNet Challenge 2019 (Sepsis Prediction) open-access dataset, along with supplementary high-fidelity and testing streams from VitalDB (2020) and the EDT-Datasets (2021), Phase 2 will aggressively focus on data preprocessing, missing value imputation, and constructing sequential time-windows. The team will establish the foundational generic FedAvg baseline locally, and subsequently train the custom decoupled FPDAF pipeline across dynamically simulated hospital nodes using PyTorch. Finally, rigorous benchmarking will be conducted, measuring drift recovery delays alongside formal evaluation metrics including Accuracy, Precision, Recall, F1-Score, and AUC-ROC to definitively quantify the optimization & personalization gains.

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